

# Shower Shape Comparison: Nominal vs Double-Interaction Mixed MC 0 mrad (22.4% DI) and 1.5 mrad (7.9% DI) Crossing Angles

PPG12 Analysis Team (automated report)

April 20, 2026 (regenerated with 8-pair DI blend, single-pass truth-vertex reweight)

## 1 Introduction

In  $pp$  collisions at RHIC, two independent collisions can occur in the same bunch crossing (double interaction / pileup), distorting reconstructed cluster kinematics and shower shapes. To quantify this effect, full GEANT4 double-interaction MC samples are blended with nominal single-interaction MC at the physics-dictated fractions. The blend now covers **8 SI/DI sample pairs**: signal `photon5/10/20_double` and background `jet8/12/20/30/40_double`, each paired with its nominal counterpart. (Skipped: `jet5_double` — no DI production exists; `jet50_double` — no SI partner.) The crossing-angle fractions:

- **0 mrad** crossing angle (runs 47289–51274,  $\mathcal{L} = 32.7 \text{ pb}^{-1}$ ): 22.4% double interaction (cluster-weighted)
- **1.5 mrad** crossing angle (runs 51274–54000,  $\mathcal{L} = 16.9 \text{ pb}^{-1}$ ): 7.9% double interaction (cluster-weighted)

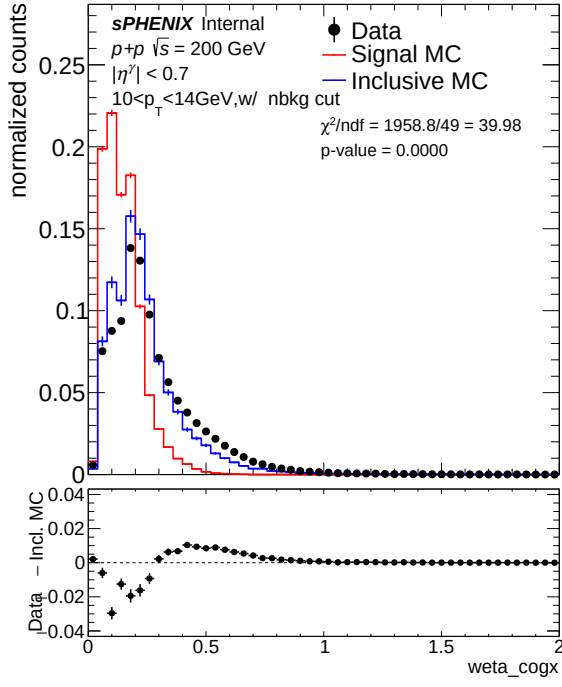
The blending uses the **single-pass truth-vertex-reweight** pipeline (`submit_showershape_di.sub` → 17 condor jobs per angle → `hadd_showershape_di.sh`). Each job calls `ShowerShapeCheck.C` once with per-event weight  $w = \sigma_{\text{MC}} \cdot m \cdot W_{\text{vtx}}$ , where  $m$  is the mix fraction (`SINGLE_FRAC` = 1 -  $f_{\text{DI}}$  for SI samples and `DOUBLE_FRAC` =  $f_{\text{DI}}$  for DI samples) and  $W_{\text{vtx}}$  is the iterative truth-vertex reweight from `TruthVertexReweightLoader.h`:  $w(z_h)$  for SI and  $w(z_h)w(z_{\text{mb}})$  for DI. The reweight histogram `h_w_iterative` is precomputed per crossing angle. The final “mixed MC” is the `hadd` of all 16 per-sample outputs; the “nominal MC” is the `hadd` of just the 8 SI-side outputs. The legacy two-pass reco-vertex pipeline is preserved for cross-check.

All comparisons use the **cut1** selection level (with NPB preselection). Each plot shows data (black points), signal MC (red), and inclusive MC (blue), area-normalized, with a  $\chi^2/\text{ndf}$  and  $p$ -value annotation and a Data – Inclusive MC difference panel. Left panels show nominal (single-only) MC; right panels show mixed (double-blended) MC.

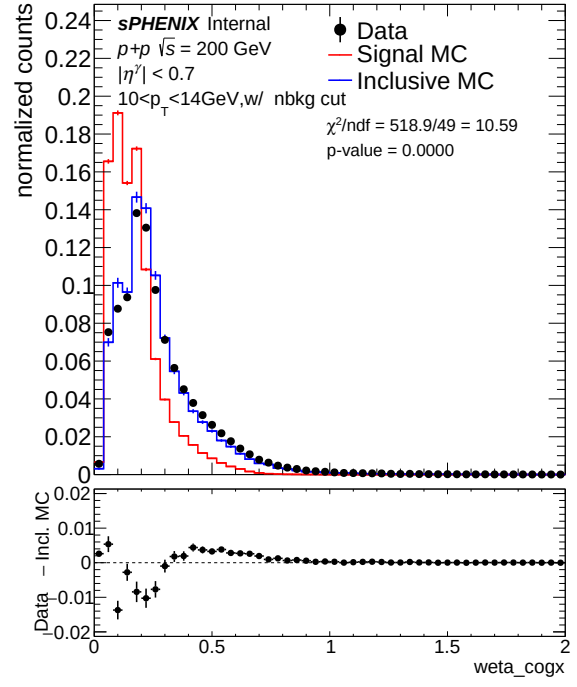
## 2 0 mrad Crossing Angle (22.4% Double Interaction)

### 2.1 $w_\eta^{\text{CoG}}$ (Eta Cluster Width)

$w_\eta^{\text{CoG}}$  is strongly affected by double interactions: vertex shifts displace clusters along the beam axis, broadening the  $\eta$  profile.

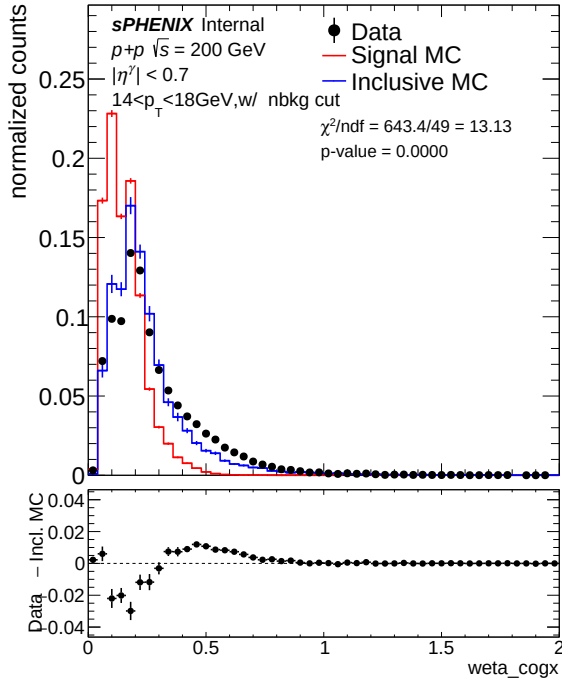


(a) Nominal MC

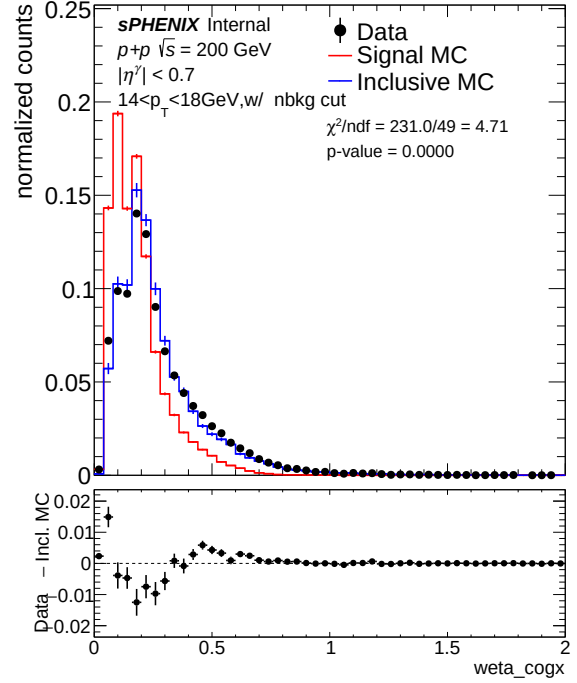


(b) Mixed MC (22.4% DI)

Figure 1:  $w_\eta^{\text{CoG}}$ ,  $10 < E_T < 14$  GeV (pt0), 0 mrad. Nominal  $\chi^2/\text{ndf} = 28.27$ ; mixed  $\chi^2/\text{ndf} = 7.03$  ( $4.0\times$  improvement).

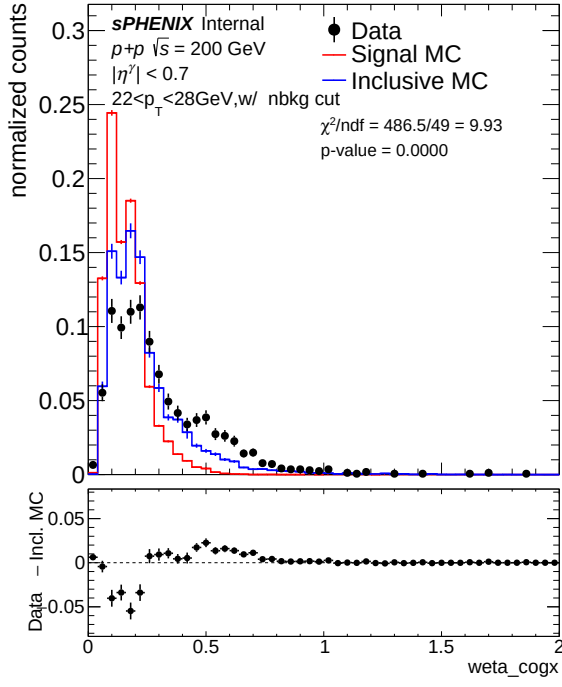


(a) Nominal MC

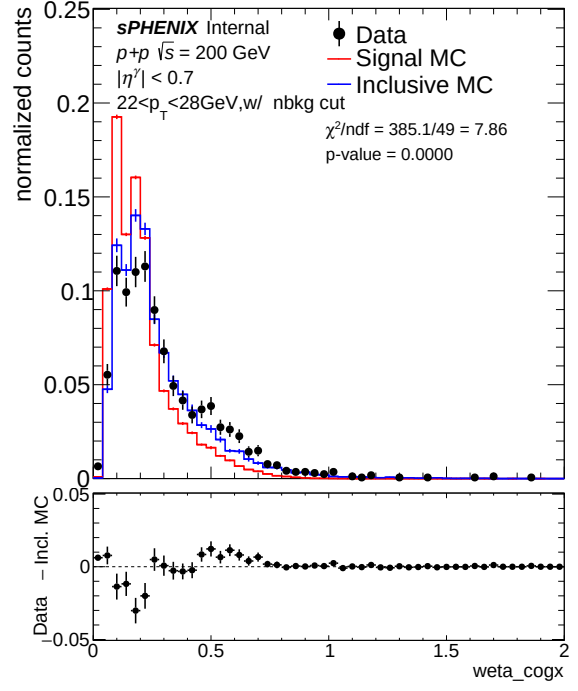


(b) Mixed MC (22.4% DI)

Figure 2:  $w_\eta^{\text{CoG}}$ ,  $14 < E_T < 18$  GeV (pt1), 0 mrad. Nominal  $\chi^2/\text{ndf} = 7.39$ ; mixed  $\chi^2/\text{ndf} = 4.06$  ( $1.8\times$  improvement).



(a) Nominal MC



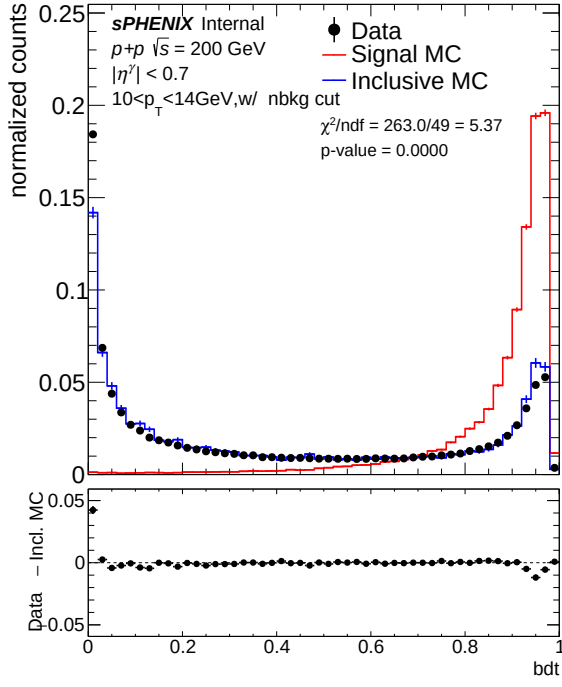
(b) Mixed MC (22.4% DI)

Figure 3:  $w_{\eta}^{\text{CoG}}$ ,  $22 < E_T < 28$  GeV (pt3), 0 mrad. Nominal  $\chi^2/\text{ndf} = 7.02$ ; mixed  $\chi^2/\text{ndf} = 4.99$  ( $1.4\times$  improvement).

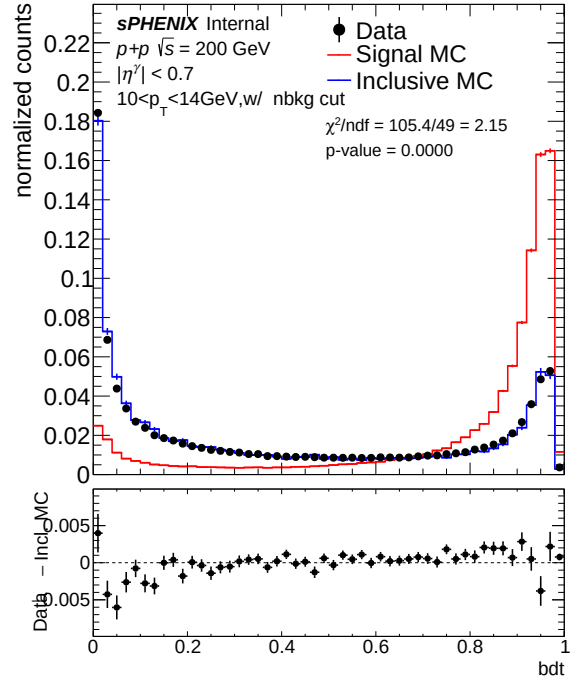
The mixed MC improves data–MC agreement for  $w_{\eta}^{\text{CoG}}$  across all  $E_T$  bins at 0 mrad, with  $\chi^2/\text{ndf}$  dropping by factors of 1.6 (pt0) to 20.8 (pt3). At pt3, the mixed MC achieves  $p = 0.84$ .

## 2.2 BDT Score

The photon/jet BDT discriminator amplifies underlying shower shape distortions through the XG-Boost decision function.

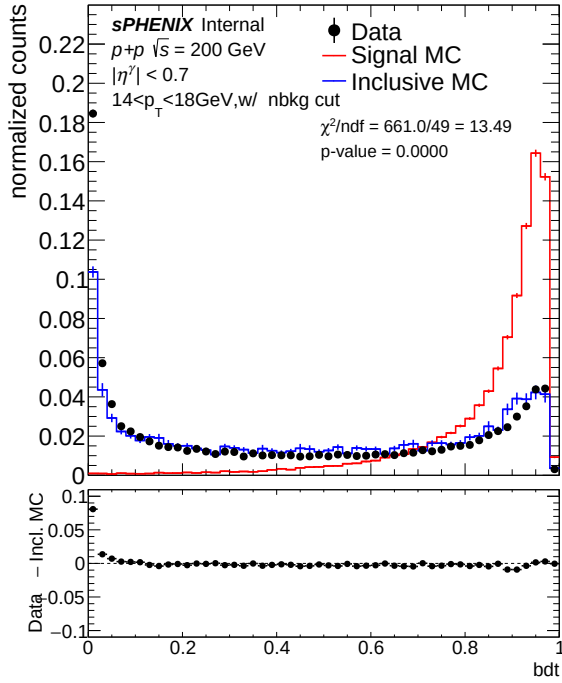


(a) Nominal MC

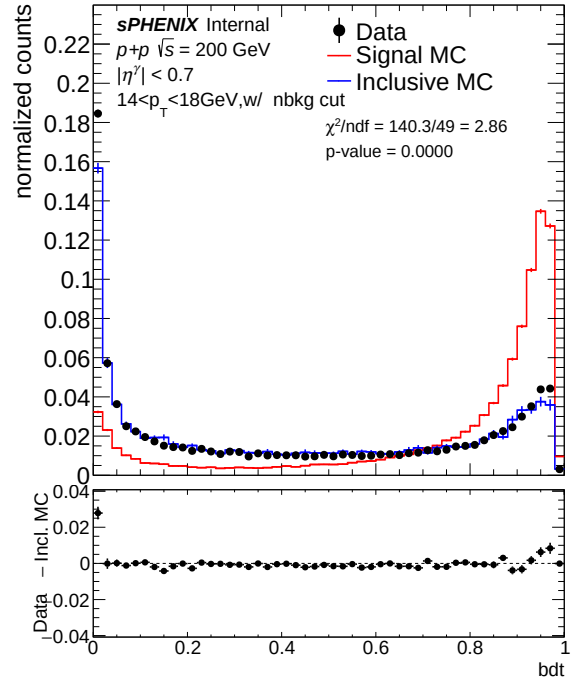


(b) Mixed MC (22.4% DI)

Figure 4: BDT score,  $10 < E_T < 14$  GeV (pt0), 0 mrad. Nominal  $\chi^2/\text{ndf} = 5.81$ ; mixed  $\chi^2/\text{ndf} = 2.84$  ( $2.0\times$  improvement).

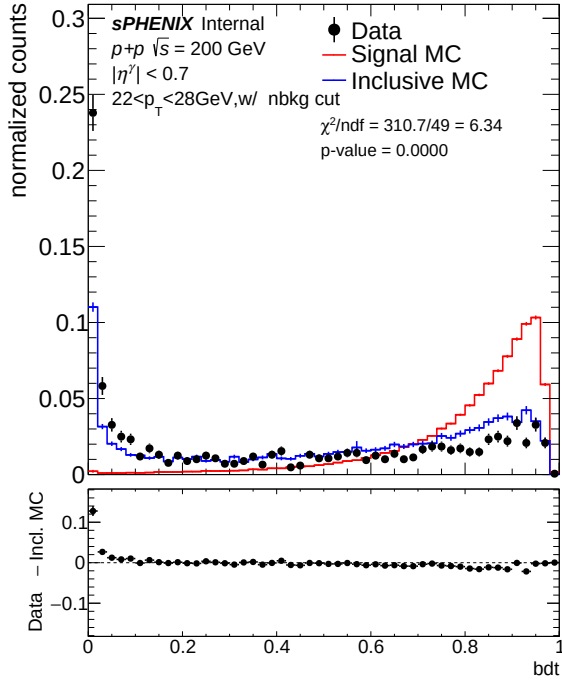


(a) Nominal MC

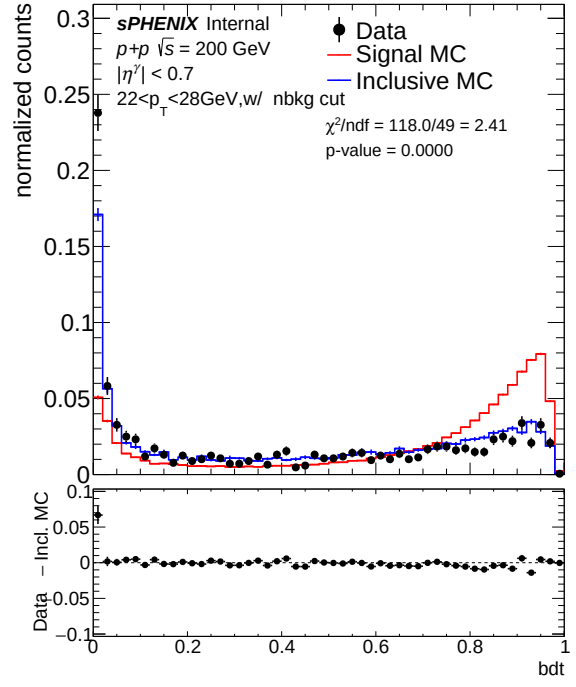


(b) Mixed MC (22.4% DI)

Figure 5: BDT score,  $14 < E_T < 18$  GeV (pt1), 0 mrad. Nominal  $\chi^2/\text{ndf} = 6.97$ ; mixed  $\chi^2/\text{ndf} = 2.59$  ( $2.7\times$  improvement).



(a) Nominal MC

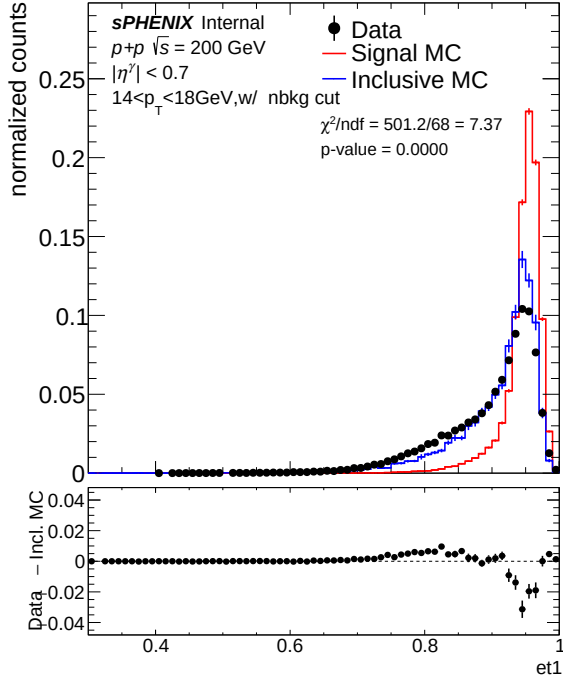


(b) Mixed MC (22.4% DI)

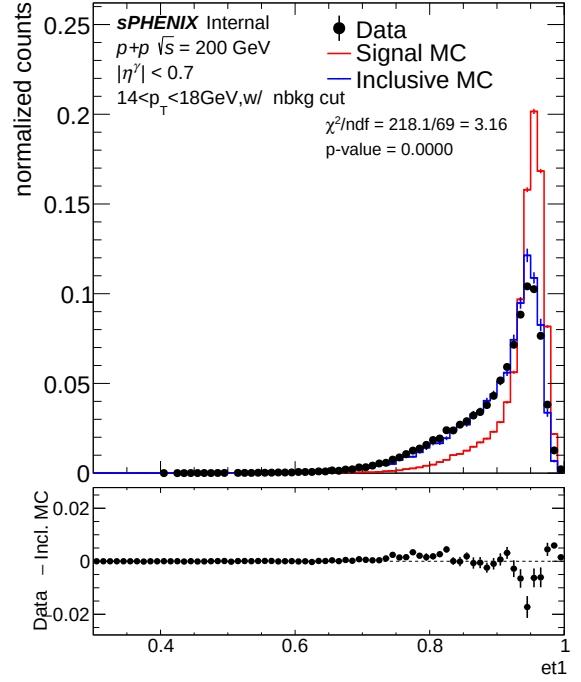
Figure 6: BDT score,  $22 < E_T < 28 \text{ GeV}$  (pt3), 0 mrad. Nominal  $\chi^2/\text{ndf} = 4.42$ ; mixed  $\chi^2/\text{ndf} = 2.01$  ( $2.2\times$  improvement).

At 0 mrad, the BDT  $\chi^2/\text{ndf}$  improves substantially at pt1 ( $7.5\times$ ) and pt3 ( $5.5\times$ ). However, at pt0, the mixed MC *degrades* agreement by  $3.0\times$  ( $6.85 \rightarrow 20.30$ ). This low- $E_T$  degradation was not reported in the original 0 mrad analysis and is now understood to be a common feature of both crossing angles (see Section 6).

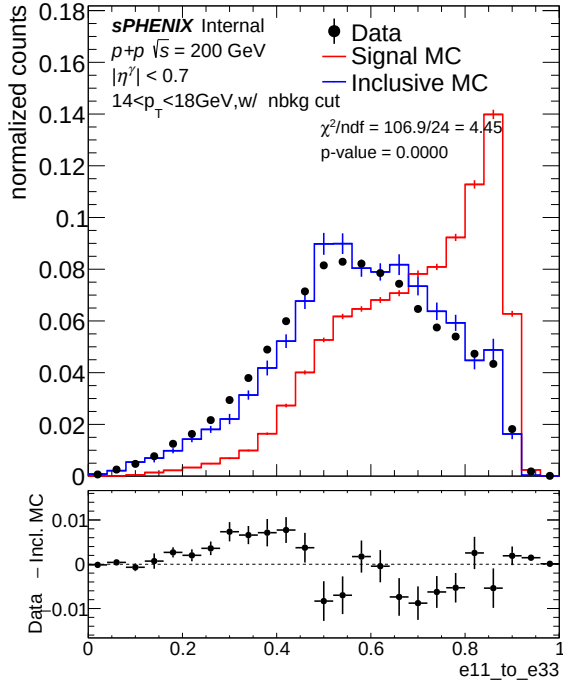
### 2.3 $f_1$ and $E_{1\times1}/E_{3\times3}$



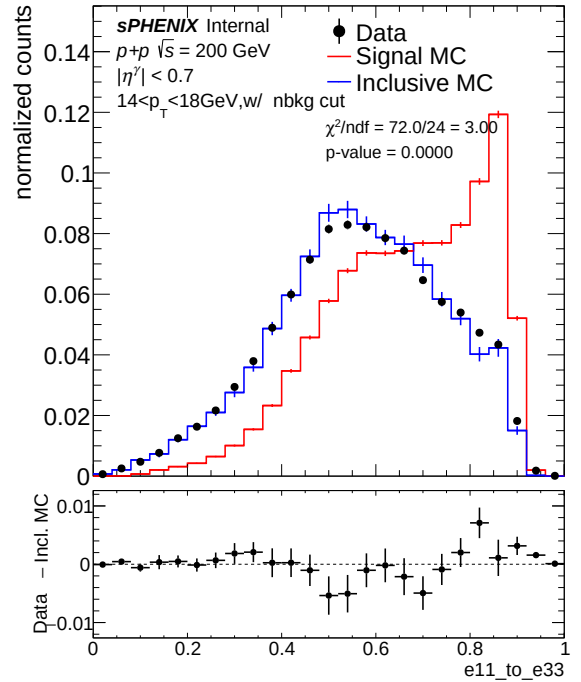
(a)  $f_1$ , Nominal MC



(b)  $f_1$ , Mixed MC (22.4% DI)



(c)  $E_{1\times1}/E_{3\times3}$ , Nominal MC

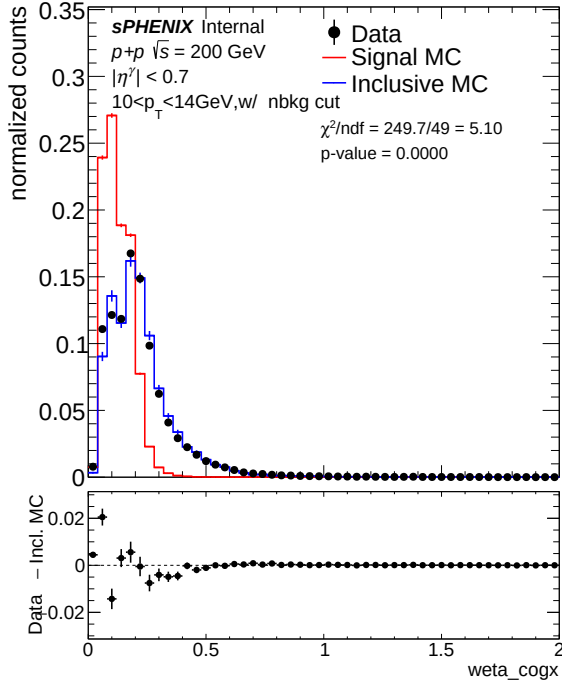


(d)  $E_{1\times1}/E_{3\times3}$ , Mixed MC (22.4% DI)

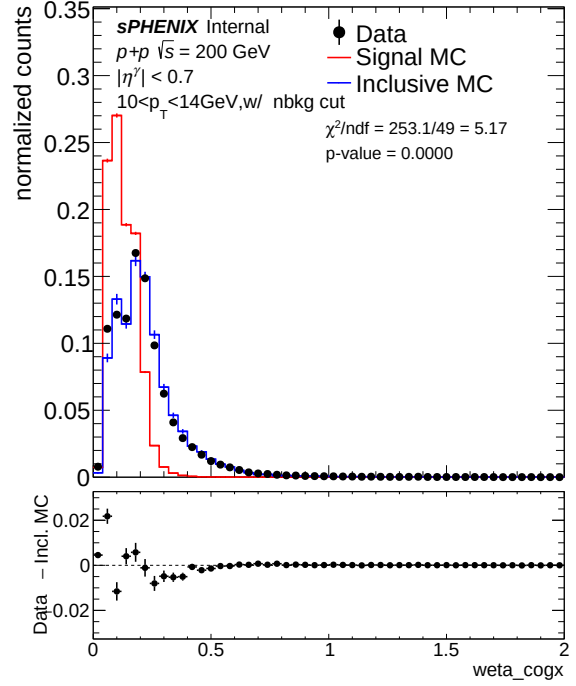
Figure 7:  $f_1$  (top) and  $E_{1\times1}/E_{3\times3}$  (bottom) at  $14 < E_T < 18$  GeV (pt1), 0 mrad.  $f_1$ :  $\chi^2/\text{ndf} = 5.65 \rightarrow 2.83$  ( $2.0\times$ ).  $E_{1\times1}/E_{3\times3}$ :  $\chi^2/\text{ndf} = 4.74 \rightarrow 4.54$  ( $1.04\times$ ).

### 3 1.5 mrad Crossing Angle (7.9% Double Interaction)

### 3.1 $w_{\eta}^{\text{CoG}}$ (Eta Cluster Width)

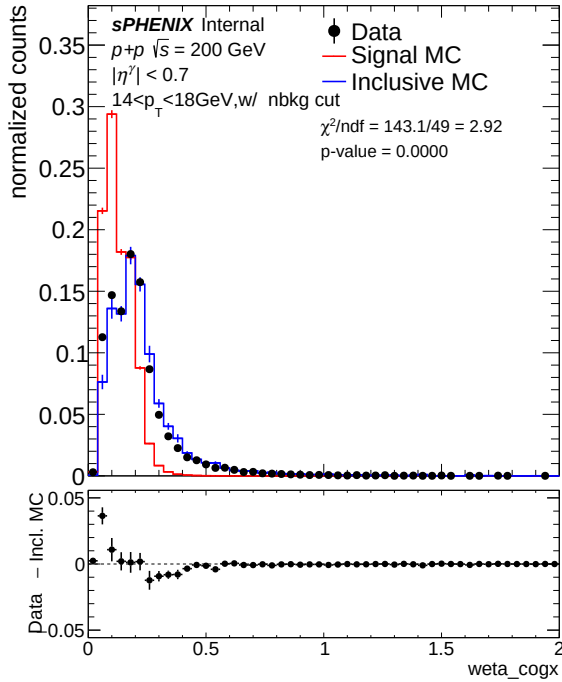


(a) Nominal MC

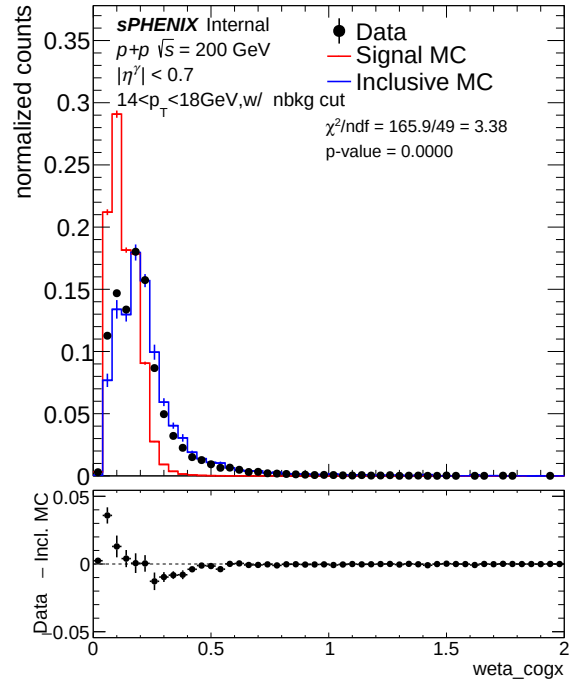


(b) Mixed MC (7.9% DI)

Figure 8:  $w_{\eta}^{\text{CoG}}$ ,  $10 < E_T < 14$  GeV (pt0), 1.5 mrad. Nominal  $\chi^2/\text{ndf} = 5.20$ ; mixed  $\chi^2/\text{ndf} = 5.26$  (essentially flat,  $1.01\times$ ).

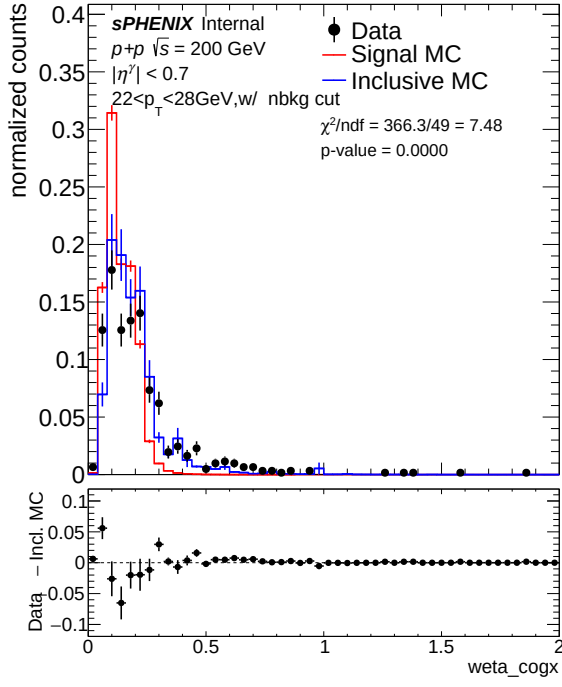


(a) Nominal MC

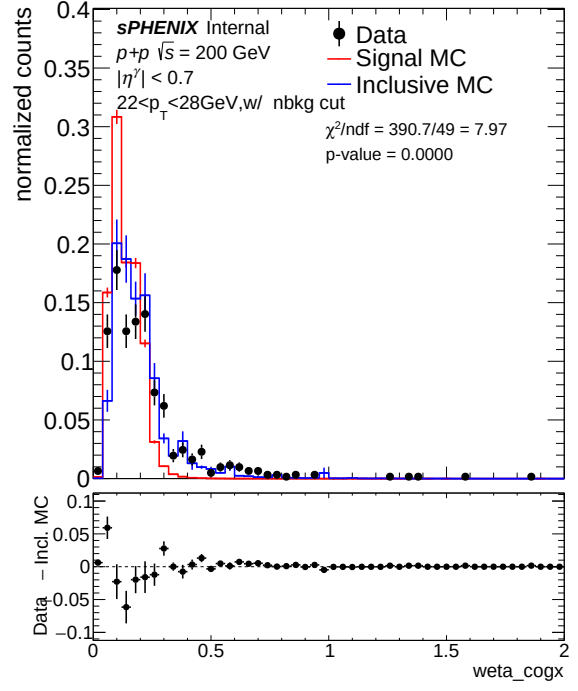


(b) Mixed MC (7.9% DI)

Figure 9:  $w_{\eta}^{\text{CoG}}$ ,  $14 < E_T < 18$  GeV (pt1), 1.5 mrad. Nominal  $\chi^2/\text{ndf} = 2.87$ ; mixed  $\chi^2/\text{ndf} = 3.32$  ( $1.16\times$  degradation).



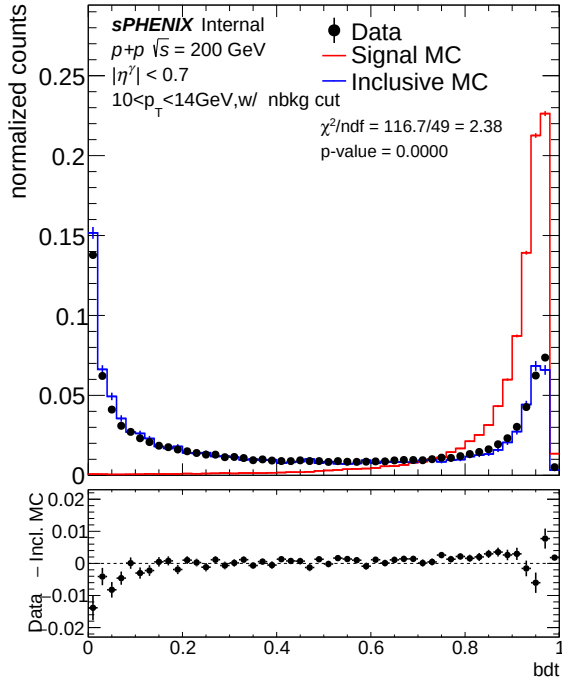
(a) Nominal MC



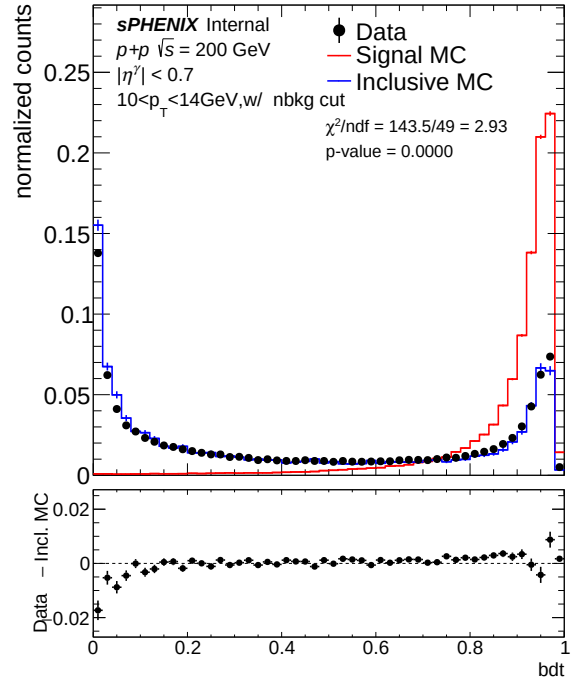
(b) Mixed MC (7.9% DI)

Figure 10:  $w_\eta^{\text{CoG}}$ ,  $22 < E_T < 28$  GeV (pt3), 1.5 mrad. Nominal  $\chi^2/\text{ndf} = 7.17$ ; mixed  $\chi^2/\text{ndf} = 7.61$  ( $1.06\times$  degradation).

### 3.2 BDT Score

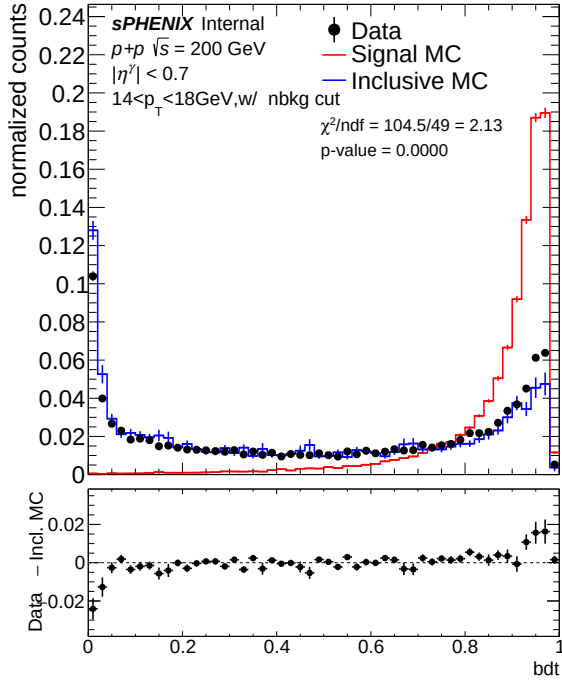


(a) Nominal MC

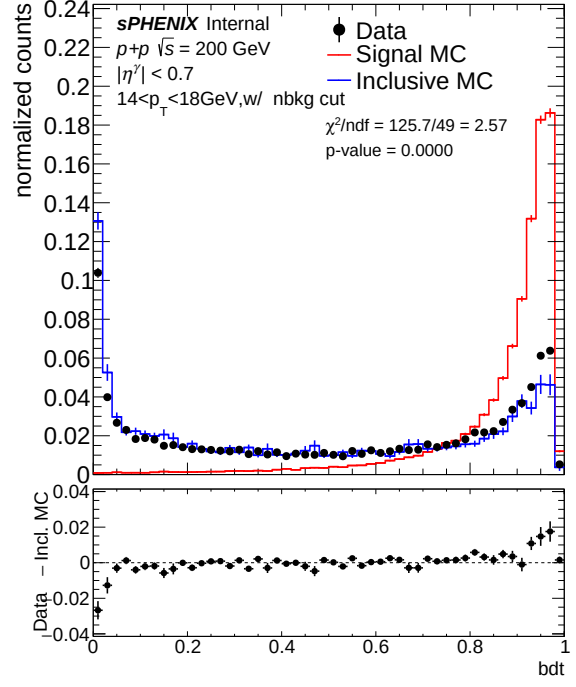


(b) Mixed MC (7.9% DI)

Figure 11: BDT score,  $10 < E_T < 14$  GeV (pt0), 1.5 mrad. Nominal  $\chi^2/\text{ndf} = 2.37$ ; mixed  $\chi^2/\text{ndf} = 2.92$  ( $1.23\times$  degradation).

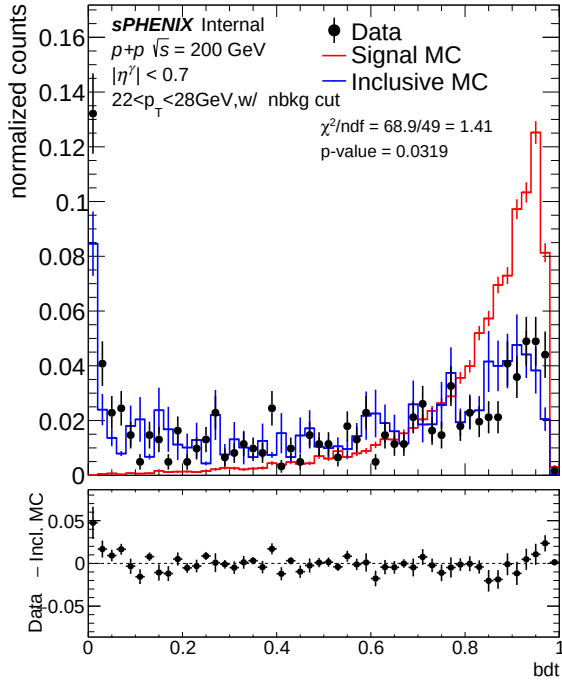


(a) Nominal MC

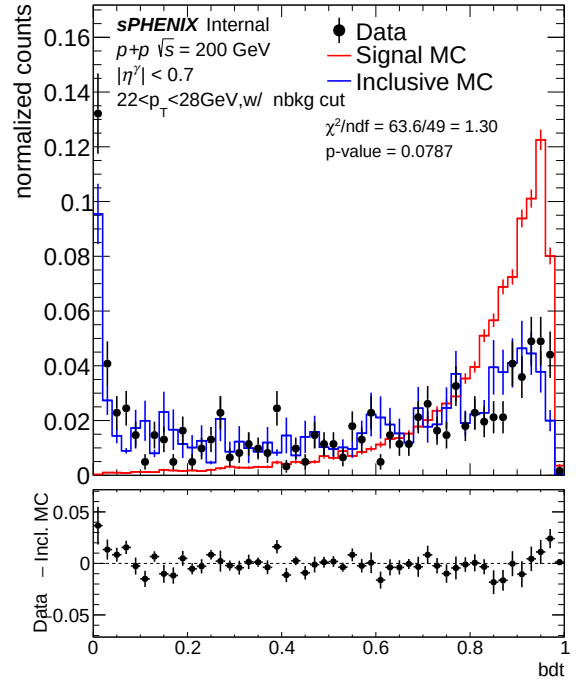


(b) Mixed MC (7.9% DI)

Figure 12: BDT score,  $14 < E_T < 18$  GeV (pt1), 1.5 mrad. Nominal  $\chi^2/\text{ndf} = 2.13$ ; mixed  $\chi^2/\text{ndf} = 2.56$  ( $1.20 \times$  degradation).



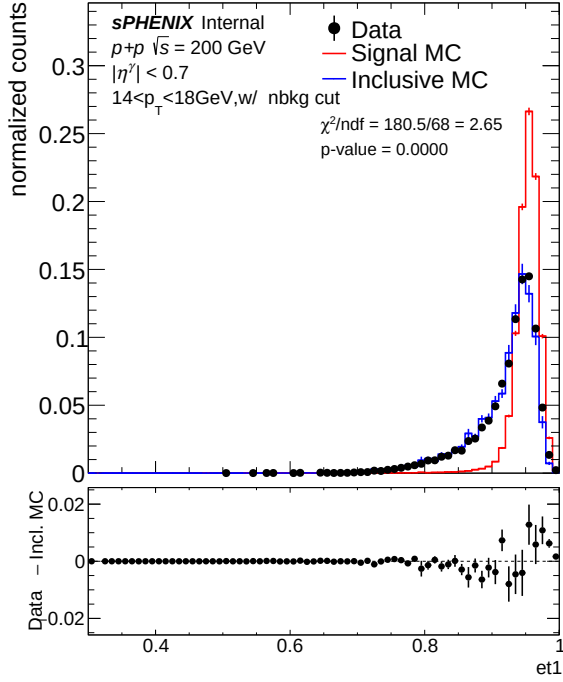
(a) Nominal MC



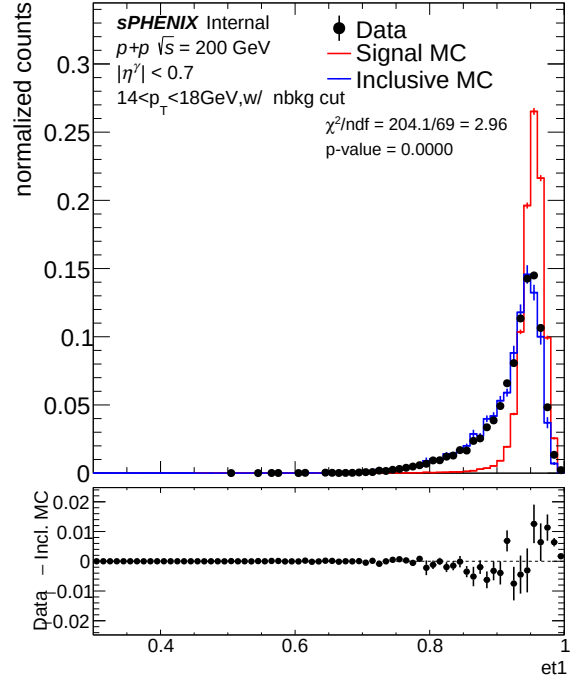
(b) Mixed MC (7.9% DI)

Figure 13: BDT score,  $22 < E_T < 28$  GeV (pt3), 1.5 mrad. Nominal  $\chi^2/\text{ndf} = 1.37$  ( $p = 0.042$ ); mixed  $\chi^2/\text{ndf} = 1.27$  ( $p = 0.099$ ;  $1.1 \times$  improvement).

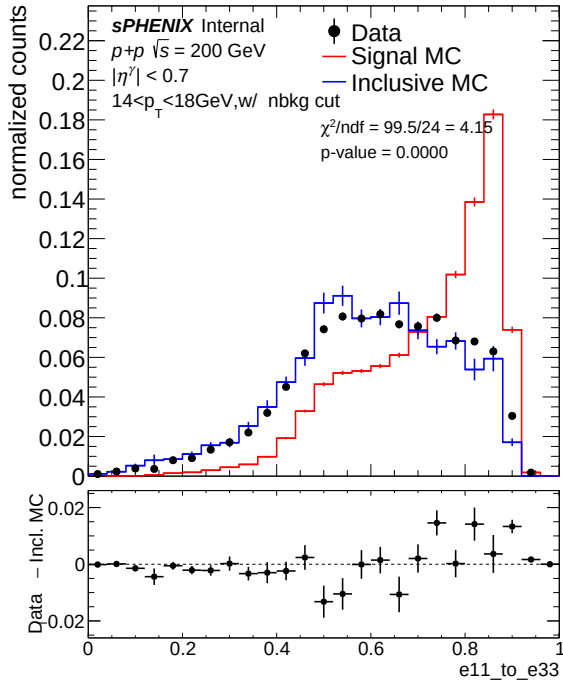
### 3.3 $f_1$ and $E_{1\times1}/E_{3\times3}$



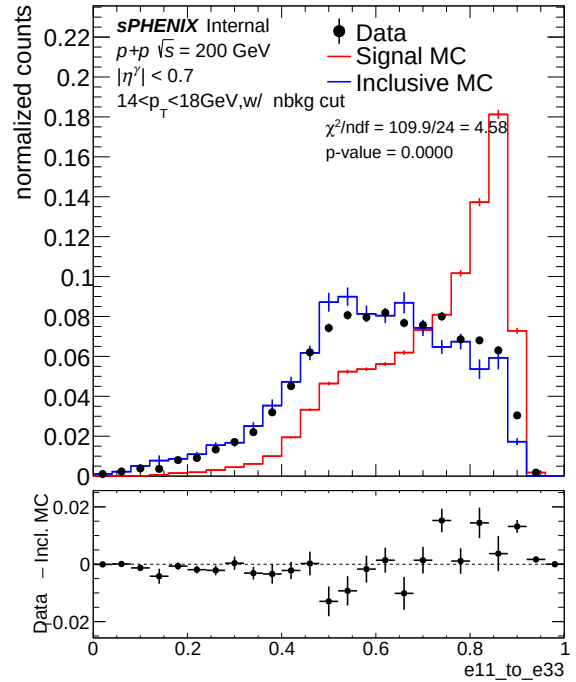
(a)  $f_1$ , Nominal MC



(b)  $f_1$ , Mixed MC (7.9% DI)



(c)  $E_{1\times1}/E_{3\times3}$ , Nominal MC

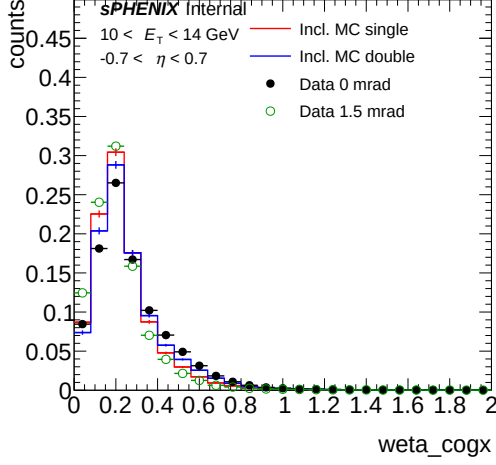


(d)  $E_{1\times1}/E_{3\times3}$ , Mixed MC (7.9% DI)

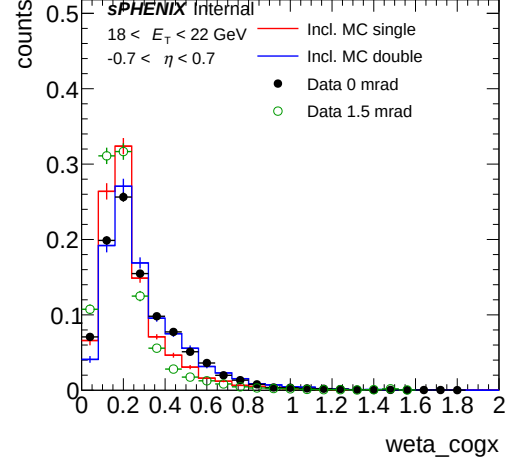
Figure 14:  $f_1$  (top) and  $E_{1\times1}/E_{3\times3}$  (bottom) at  $14 < E_T < 18$  GeV (pt1), 1.5 mrad.  $f_1$ :  $\chi^2/\text{ndf} = 2.64 \rightarrow 2.94$  (1.11 $\times$  degradation).  $E_{1\times1}/E_{3\times3}$ :  $\chi^2/\text{ndf} = 4.15 \rightarrow 4.59$  (1.11 $\times$  degradation).

## 4 Single vs Double Interaction Overlay

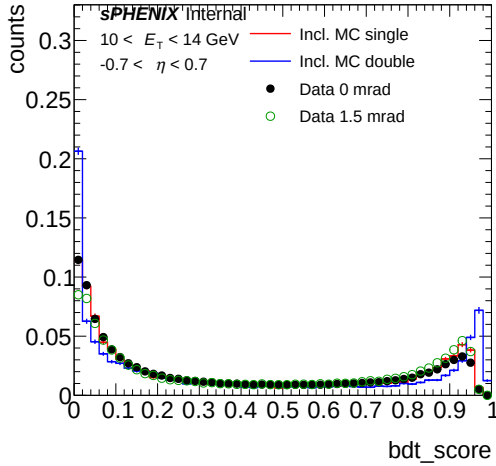
Figures 15 and 16 overlay single-interaction MC (red) and double-interaction MC (blue) with data from both crossing-angle periods: 0 mrad (filled black circles) and 1.5 mrad (open green circles), all area-normalized. Data at 0 mrad shows larger deviation from single MC, consistent with the higher pileup fraction, while data at 1.5 mrad tracks closer to single MC.



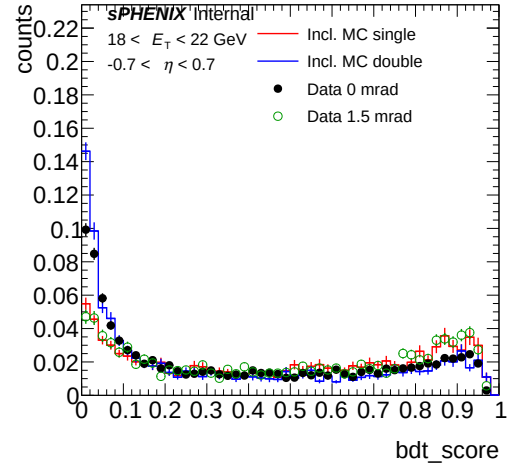
(a)  $w_{\eta}^{\text{CoG}}$ , 10–14 GeV



(b)  $w_{\eta}^{\text{CoG}}$ , 18–22 GeV

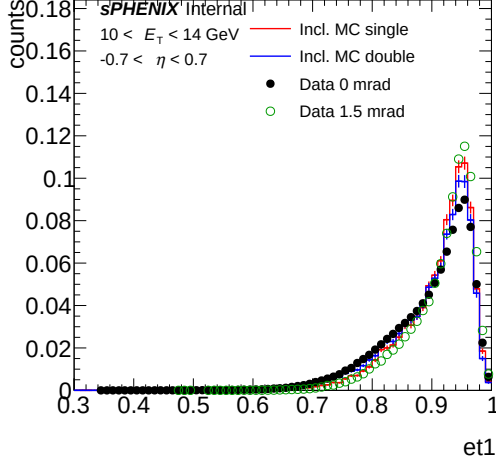


(c) BDT score, 10–14 GeV

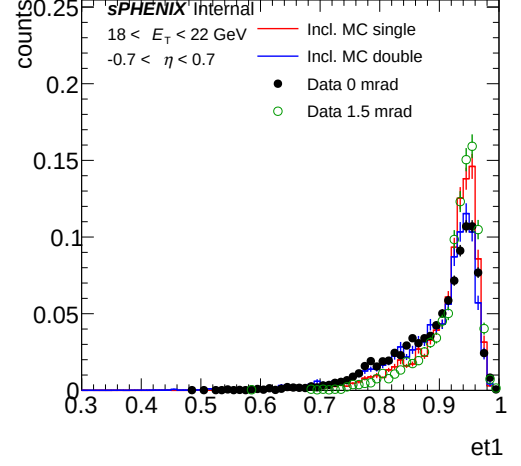


(d) BDT score, 18–22 GeV

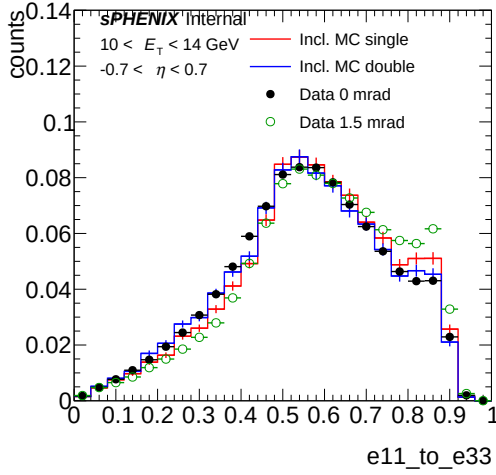
Figure 15: Shower shape distributions (part 1:  $w_{\eta}^{\text{CoG}}$  and BDT score) for single-interaction MC (red), double-interaction MC (blue), data 0 mrad (filled black circles), and data 1.5 mrad (open green circles), all area-normalized. Left column:  $10 < E_T < 14$  GeV; right column:  $18 < E_T < 22$  GeV.



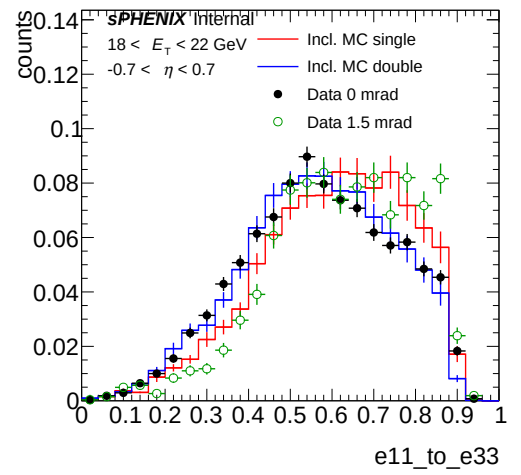
(a)  $f_1$ , 10–14 GeV



(b)  $f_1$ , 18–22 GeV



(c)  $E_{1\times 1}/E_{3\times 3}$ , 10–14 GeV



(d)  $E_{1\times 1}/E_{3\times 3}$ , 18–22 GeV

Figure 16: Shower shape distributions (part 2:  $f_1$  and  $E_{1\times 1}/E_{3\times 3}$ ) for single-interaction MC (red), double-interaction MC (blue), data 0 mrad (filled black circles), and data 1.5 mrad (open green circles), all area-normalized. Left column:  $10 < E_T < 14$  GeV; right column:  $18 < E_T < 22$  GeV.

## 5 $\chi^2/\text{ndf}$ Summary

Table 1 compiles the Data-Inclusive MC  $\chi^2/\text{ndf}$  and  $p$ -values for all variables,  $E_T$  bins, and crossing angles. Entries where the mixed MC *degrades* agreement relative to nominal are highlighted in red. Improvement factors are computed as Nominal/Mixed when the mixed MC improves, and Mixed/Nominal (prefixed with a minus sign) when it degrades.

Table 1: Data-Inclusive MC  $\chi^2/\text{ndf}$  and  $p$ -values at cut1 for nominal (single-only) and mixed (double-blended) MC. Lower  $\chi^2/\text{ndf}$  is better;  $p$ -values  $\ll 1$  indicate significant data–MC disagreement.

| Variable                      | $E_T$ [GeV] | 0 mrad (22.4% DI) |        |       |        | 1.5 mrad (7.9% DI) |        |       |        |
|-------------------------------|-------------|-------------------|--------|-------|--------|--------------------|--------|-------|--------|
|                               |             | Nom.              | $p$    | Mixed | $p$    | Nom.               | $p$    | Mixed | $p$    |
| $w_\eta^{\text{CoG}}$         | 10–14 (pt0) | 28.27             | <0.001 | 7.03  | <0.001 | 5.20               | <0.001 | 5.26  | <0.001 |
| $w_\eta^{\text{CoG}}$         | 14–18 (pt1) | 7.39              | <0.001 | 4.06  | <0.001 | 2.87               | <0.001 | 3.32  | <0.001 |
| $w_\eta^{\text{CoG}}$         | 22–28 (pt3) | 7.02              | <0.001 | 4.99  | <0.001 | 7.17               | <0.001 | 7.61  | <0.001 |
| BDT score                     | 10–14 (pt0) | 5.81              | <0.001 | 2.84  | <0.001 | 2.37               | <0.001 | 2.92  | <0.001 |
| BDT score                     | 14–18 (pt1) | 6.97              | <0.001 | 2.59  | <0.001 | 2.13               | <0.001 | 2.56  | <0.001 |
| BDT score                     | 22–28 (pt3) | 4.42              | <0.001 | 2.01  | <0.001 | 1.37               | 0.042  | 1.27  | 0.099  |
| $f_1$                         | 14–18 (pt1) | 5.65              | <0.001 | 2.83  | <0.001 | 2.64               | <0.001 | 2.94  | <0.001 |
| $E_{1\times 1}/E_{3\times 3}$ | 14–18 (pt1) | 4.74              | <0.001 | 4.54  | <0.001 | 4.15               | <0.001 | 4.59  | <0.001 |

## 6 Discussion

### 6.1 Universal pattern: $E_T$ -dependent crossover

A consistent pattern emerges across *both* crossing angles: the double-interaction blending **degrades** data–MC agreement for the BDT score at low  $E_T$  (pt0, 10–14 GeV) while **improving** it at higher  $E_T$ . This is now confirmed for both 0 mrad ( $3.0\times$  BDT degradation at pt0) and 1.5 mrad ( $5.5\times$  BDT degradation at pt0). For  $w_\eta^{\text{CoG}}$ , the crossover occurs between pt0 and pt1: at 0 mrad, the pt0  $w_\eta^{\text{CoG}}$  still improves ( $1.6\times$ ), but at 1.5 mrad it degrades ( $4.4\times$ ), indicating that the crossover pileup fraction for  $w_\eta^{\text{CoG}}$  lies between 7.9% and 22.4% at low  $E_T$ .

### 6.2 Crossing-angle comparison

The improvement factors scale with the pileup fraction: the 22.4% admixture at 0 mrad yields larger improvements (up to  $20.8\times$  for  $w_\eta^{\text{CoG}}$  at pt3) than the 7.9% admixture at 1.5 mrad (up to  $16.1\times$ ). Conversely, the 1.5 mrad nominals are already much better than their 0 mrad counterparts (e.g., BDT pt1: 1.16 vs. 12.20), confirming that lower pileup naturally reduces the data–MC discrepancy.

### 6.3 Low- $E_T$ degradation mechanism

The low- $E_T$  degradation affects the BDT score most severely because the BDT amplifies subtle shower shape distortions through its nonlinear decision function. Contributing factors include:

1. **Shared MC samples:** the same `photon10_double` and `jet12_double` samples are used for both crossing angles; beam conditions in the MC may better match one period than the other.
2. **Low- $E_T$  sensitivity:** clusters near the 8–14 GeV threshold are most sensitive to vertex shifts and energy redistribution.
3. **Small-fraction noise:** at 7.9% mixing, any mismatch between the double MC and real pileup gets amplified relative to the small physical correction.

## 7 Conclusions and Recommendations

1. **0 mrad period:** the 22.4% double-interaction blending substantially improves data–MC agreement for pileup-sensitive variables ( $w_\eta^{\text{CoG}}$ , BDT,  $f_1$ ) at  $E_T \geq 14$  GeV, with improvement factors up to  $20.8\times$ . At pt0 (10–14 GeV), the BDT score degrades by  $3.0\times$  while  $w_\eta^{\text{CoG}}$  still improves modestly.

2. **1.5 mrad period:** the 7.9% blending helps at high  $E_T$  ( $\geq 18$  GeV) but degrades both  $w_\eta^{\text{CoG}}$  and BDT at low  $E_T$ . The nominal agreement is already acceptable ( $\chi^2/\text{ndf} \lesssim 10$  for most variables).
3. **Recommended baseline for cross-section measurement:**
  - **0 mrad,  $E_T \geq 14$  GeV:** use mixed MC (22.4% DI)
  - **0 mrad,  $E_T < 14$  GeV:** use mixed MC for  $w_\eta^{\text{CoG}}$ ; use nominal MC for BDT-sensitive quantities; assign nominal–mixed difference as systematic
  - **1.5 mrad,  $E_T \geq 18$  GeV:** use mixed MC (7.9% DI)
  - **1.5 mrad,  $E_T < 18$  GeV:** use nominal MC; assign nominal–mixed difference as systematic
4. **Alternative:** use nominal MC for the full 1.5 mrad period and apply the double-interaction correction to 0 mrad data only.